Minh-Thien Nguyen

AI Researcher

Ho Chi Minh City | minhnguyent546@gmail.com | (+84) 373 470 203

Portfolio | in/minhnguyent546 | minhnguyent546

EDUCATION

Engineer's degree, Information Technology **08/2021 - 11/2025**

Can Tho University **GPA:** 3.91/4.0

- **Focus:** Deep Learning, Natural Language Processing, Large Language Models, Computer Vision.
- Ranked **Top 3 GPA** among Information Technology majors.
- **Thesis:** Building a Smart Enrollment Advisory System Based on an Automatic Dialogue Model.
- **Thesis Grade:** 9.6/10.

PUBLICATIONS

Quoc-Khang Tran*, **Minh-Thien Nguyen***, Phu-An Thai, Xuan-Tung Bui, Truong-Thanh Ma, and Nguyen-Khang Pham. "CoTu at EXACT 2026: Neuro-Symbolic Reasoning for Transparent Educational QA". **Accepted** at the *15th International Conference on Computational Science and Network Intelligence (**CSoNet 2026**)*, *Special Session: EXACT Competition*, Springer LNCS, 2026 (arXiv).

- Built a **neuro-symbolic Program-of-Thought** system where a **4B** open-weight backbone writes a program instead of answering directly: **Z3** constraints for logical queries over university regulations, numerical Python for multi-step physics, both sandboxed with a self-correction loop.
- Placed **3rd of 50 teams** under an **≤8B** open-weight, self-hosted, explain-every-answer constraint, with a **perfect physics score** in both selection rounds and the **highest technical score** of any team in the final (**13.44**/15 vs. 12.51 runner-up), scored on automated answer accuracy plus **expert-judged reasoning depth**.
- Optimized serving to meet a **60 s** per-query limit: replaced **vLLM** with **SGLang** and added **DFLASH** speculative decoding, cutting mean latency to 12.1 s (logic) and 4.2 s (physics) while running the thinking backbone unquantized in BF16.

Quoc-Khang Tran*, **Minh-Thien Nguyen***, and Nguyen-Khang Pham. "ViCLIP-OT: A Structure-Aware Vision-Language Model for Vietnamese Image-Text Retrieval". **Under review** at *Machine Vision and Applications*, 2026 (arXiv).

- Introduced **ViCLIP-OT**, a vision-language model for Vietnamese image–text retrieval that combines CLIP-style contrastive learning with a **Similarity-Graph Regularized Optimal Transport (SIGROT)** loss.
- Outperformed CLIP and SigLIP baselines on three Vietnamese benchmarks (UIT-OpenViIC, KTVIC, Crossmodal-3600): **67.34%** average Recall@K on UIT-OpenViIC (**+5.75** pp over CLIP), and **+11.72** pp over CLIP zero-shot on Crossmodal-3600.
- Confirmed through embedding-space analysis that SIGROT improves cross-modal alignment and **reduces the modality gap**.

Quoc-Khang Tran, **Minh-Thien Nguyen**, and Nguyen-Khang Pham. "Leveraging Model Soups to Classify Intangible Cultural Heritage Images from the Mekong Delta". *Journal on Information Technologies & Communications (**ICT Research**)*. Proceedings of the International Conference on Intelligent Systems and Data Science (ISDS), Oct 2025 (arXiv).

- Proposed a framework that combines the hybrid **CoAtNet** architecture with **model soups** to classify Intangible Cultural Heritage (ICH) images from the Mekong Delta.
- Reached **72.36%** top-1 accuracy and **69.28%** macro F1 on the ICH-17 dataset (7,406 images across 17 classes), ahead of ResNet-50, DenseNet-121, ViT, and earlier studies on the same dataset.
- Explained the gain through **bias–variance decomposition**, showing model soups reduces variance while adding little bias, and used cross-entropy distance metrics with **multidimensional scaling** to show it selects

geometrically diverse checkpoints where Soft Voting blends redundant ones.

*Equal contribution.

EXPERIENCE

Independent AI Researcher **08/2025 - Present**

Independent Research · In collaboration with CICT, Can Tho University

- Research **inference-time scaling** for LLM reasoning, **embedding models**, **Vietnamese image–text retrieval**, and **optimal transport**: set the research questions; design, implement, and run the experiments; and write the papers with co-authors and an academic advisor.
- Produced **three papers** to date — one published (**ICT Research**, ISDS 2025), one accepted (**CSoNet 2026**), and one under review (*Machine Vision and Applications*) — and a **3rd-place finish among 50 teams** in the **EXACT 2026** challenge at IEEE IJCNN 2026 (see **Publications**).

Data Science and AI Intern **05/2026 - 08/2026**

Viettel AI (Selected via Viettel Digital Talent 2026 program)

- Researched prompting and decoding strategies to improve LLM performance on multi-step reasoning for Vietnamese benchmarks.
- Curated a **294-item Vietnamese multi-step logical-reasoning dataset**: translated an English item pool with TranslateGemma-27B, then re-annotated labels with a **Symbolic Reasoning** pipeline (program-of-thought plus a Z3 solver, via DeepSeek-V4-Pro). Validated the labels against 50 human-annotated items at **96.7%** agreement (answer accuracy and supporting-premise F1).
- Built **ViREx-Bench**, an **inference-time scaling** evaluation framework supporting 7 reasoning strategies on any OpenAI-compatible model (CoT, self-consistency, Tree-of-Thought (Beam, DFS, and MCTS), Cumulative Reasoning, and the proposed **Symbolic Reasoning pipeline**); benchmarked against 8 models.
- Found CoT captures most of the accuracy gain, while search-based strategies use up to 32x more tokens without beating it; the proposed **Symbolic Reasoning pipeline** led the search-based strategies in accuracy at **3.3 model calls per item**, versus 18–28 for search-based ones.
- Framework: minhnguyent546/ViREx-Bench · Technical report: Link.

AI Engineer **06/2025 - 08/2025**

CT Group Corporation

- Selected for the role through a competitive talent program run jointly by **CT Group Corporation** and **Can Tho University**.
- Evaluated large language models (**ChatGPT**, **Grok**, **Gemini**) and AI agent frameworks (**ManusAI**, OpenManus, AgenticSeek) for integration into existing systems.
- Built a **RAG**-based chatbot with **FastAPI and Qdrant** that reviews candidate information and recommends interview departments.

AI Engineer Intern **12/2024 - 04/2025**

TMA Solutions

- Researched 3D human body modeling and facial expression analysis methods to design the project's technical approach.
- Built a model that generates a 3D human body from a user's height and weight, using **SMPL** and **SMPL-X** models with the Open3D library for parameter computation and rendering.
- Built a **facial expression analysis** module for an AI system that supports autistic children.

PROJECTS

Multiple-Choice Question Answering (MCQA) Pipeline for Complex Technical Documents — *Viettel AI Race 2025* (github) **09/2025 - 11/2025**

- Built an end-to-end pipeline for MCQA over technical PDFs that contain tables, charts, and images. Placed **Top 5** in online round 1 and **Top 3** in online rounds 2 and 3 on the public leaderboard.
- Combined a multimodal extraction layer (**DeepSeek-OCR**, **Qwen3-VL-4B-Thinking**) that scored **87.05**

on document extraction with a **RAG** retrieval stage (hybrid dense and sparse search, query expansion, neighbor expansion) and **Qwen3-4B-Thinking** for answer generation, which answered **91.2%** of the round-3 questions correctly.

- Served inference with **vLLM** and cut end-to-end processing time by **4x** compared to Hugging Face inference.
- **Tech stack:** RAG, DeepSeek-OCR, Qwen3, Qwen3-VL, vLLM, Qdrant, MinerU, Pydantic.

***Thesis* - Building a smart enrollment advisory system based on an automatic dialogue model (github)** **05/2025 - 08/2025**

- Built a Smart Enrollment Advisory System (SEAS), a **RAG**-based chatbot for Can Tho University, with a retrieval module on LangChain and Qdrant and an async FastAPI backend on Postgres.
- Compared **query expansion**, **reranking**, and both together against a dense-retrieval baseline. Reranking accounted for nearly all of the improvement, and lifted **MRR@10** from 0.1839 to 0.7008 and **NDCG@10** from 0.3120 to 0.7062.
- Selected query expansion plus reranking as the final configuration, which reached **0.7713** Recall@10, **0.7006** MRR@10, **0.6906** MAP@10, and **0.7133** NDCG@10, at an average response time of **4.741 seconds** with **token streaming** through Gemini 2.5 Flash.
- **Tech stack:** FastAPI, LangChain, Qdrant, SQLAlchemy, Pydantic, AWS (Aurora, EC2), ReactJS, TypeScript, Docker, Docker Compose.
- **Advisor:** Assoc. Prof. Dr. Nguyen-Khang Pham · **Thesis Grade: 9.6**/10.

Medical Llama-2 (github · demo) **08/2024 - 11/2024**

- Adapted **LLaMA 2** to the medical question-answering domain with parameter-efficient fine-tuning (**PEFT**), using **QLoRA**.
- Trained the model with distributed data parallel (**DDP**), then quantized and served it through **llama.cpp** for local inference.
- **Tech stack:** Transformers, Peft, Bitsandbytes, PyTorch, Gradio, LLama.cpp, Docker.

Pre-training GPT-2 (github) **05/2024 - 08/2024**

- Pre-trained **GPT-2** (124M parameters) from scratch on the **Fineweb-Edu 10BT** subset with a **TPUv3-8**, and reached a perplexity of ∼**21.02**, which is better than OpenAI's original GPT-2 checkpoint.
- Scaled the training run across **multi-GPU and TPU** devices with distributed data parallel through **PyTorch/XLA**.
- **Tech stack:** PyTorch, Pytorch/XLA for XLA devices (TPU and GPU), Distributed Data Parallel training (DDP).

SKILLS

- **Programming:** Python, C/C++.
- **ML/DL frameworks:** PyTorch, Transformers, PEFT, TRL, Scikit-learn.
- **Training, evaluation & efficiency:** distributed data parallel (DDP), Fully Sharded Data Parallel (FSDP), TPU/GPU training via PyTorch/XLA, QLoRA/LoRA, quantization (bitsandbytes), W&B, LM-Eval.
- **LLM serving & inference:** vLLM, SGLang, TensorRT-LLM, Llama.cpp, DFLASH speculative decoding.
- **LLM application development:** LangChain, LangGraph, FastAPI, Qdrant, SQLAlchemy, Z3 (SMT solver).
- **Infrastructure:** PostgreSQL, Docker, AWS (EC2, Aurora).

AWARDS

- Third prize at the 33rd Vietnam Student Olympiad in Informatics (Super Cup division), 2024.
- Third prize at the 2023 ICPC Vietnam National Programming Contest.
- Second prize at the 31st Vietnam Student Olympiad in Informatics (Specialized Informatics division), 2022.
- Third prize at the 2022 ICPC Vietnam National Programming Contest.
- Third prize at the 30th Vietnam Student Olympiad in Informatics (Non-specialized Informatics division), 2021.

CERTIFICATIONS

TOEIC (L&R): 845/990 (issued by the **Educational Testing Service (ETS)** on **June 2024**)

COMMUNITY ENGAGEMENT

- **Problem setter and tester** for the **CICT Collegiate Programming Contest (CICTCPC)** — *College of Information and Communication Technology, Can Tho University, 2025*. Designed original problems, wrote reference solutions, and built test data.
- **Problem setter and tester** for the **CICT selection contest** that chose the university team for the **Vietnam Student Olympiad in Informatics** — *Can Tho University, 2024 and 2025*.
- Author of **Virtual tree** — *VNOI Magazine, 2024* (magazine).
- Author of **Subtle Techniques with the XOR Operation** — *VNOI Magazine, 2023* (magazine).